The Discrete Dirichlet Problem: A Maximum Principle and Unique Solvability

Colin McConnell

Source: M. Artin, *Algebra*, Chapter 1, Miscellaneous Problem M.11 (MIT OCW 18.701, Problem Set 1)

Problem Statement. A function $f(u, v)$ on the integer lattice is *harmonic* at (u, v) if

$$f(u+1, v) + f(u-1, v) + f(u, v+1) + f(u, v-1) - 4f(u, v) = 0,$$

that is, if $f(u, v)$ is the average of its values at the four neighbors of (u, v). A *discrete region* R is a finite set of lattice points. Its *boundary* ∂R is the set of lattice points not in R that are at distance 1 from some point of R, and $\overline{R} = R \cup \partial R$. Given a function β on ∂R, the *discrete Dirichlet problem* asks for a function f on $\overline{R}$ that equals β on ∂R and is harmonic at every point of R. Writing $x_{uv} = f(u, v)$ for the unknown values at points of R, this becomes a linear system $LX = B$.

(a) Let $R = \{(0,0), (0, \pm 1), (\pm 1, 0)\}$. Write the system and solve it when $\beta_{uv} = 0$ for $v \leq 0$ and $\beta_{uv} = 1$ for $v > 0$.

(b) Prove the maximum principle for discrete harmonic functions.

(c) Prove that the discrete Dirichlet problem has a unique solution for every region R and every boundary function β.

Throughout, we assume R is nonempty.

Observation. If $p \in R$, then each of the four neighbors of p lies in $\overline{R}$. Indeed, a neighbor q is either in R, or it is not in R and is at distance 1 from $p \in R$, in which case $q \in \partial R$. So the harmonic condition at a point of R only refers to values of f on $\overline{R}$.

Part (a)

The boundary. The neighbors of each point of R are listed below. Those not in R are underlined.

Point of R	Neighbors
$(0,0)$	$(1,0),\ (-1,0),\ (0,1),\ (0,-1)$
$(1,0)$	$\underline{(2,0)},\ (0,0),\ \underline{(1,1)},\ \underline{(1,-1)}$
$(-1,0)$	$(0,0),\ \underline{(-2,0)},\ \underline{(-1,1)},\ \underline{(-1,-1)}$
$(0,1)$	$\underline{(1,1)},\ \underline{(-1,1)},\ \underline{(0,2)},\ (0,0)$
$(0,-1)$	$\underline{(1,-1)},\ \underline{(-1,-1)},\ (0,0),\ \underline{(0,-2)}$

Hence

$$\partial R = \{(\pm 2, 0),\ (0, \pm 2),\ (1,1),\ (1,-1),\ (-1,1),\ (-1,-1)\},$$

which has eight points. The boundary function is

$$\beta = 1 \text{ at } (0,2),\ (1,1),\ (-1,1), \qquad \beta = 0 \text{ at } (2,0),\ (-2,0),\ (0,-2),\ (1,-1),\ (-1,-1).$$

The system. Order the points of R as $(0,0), (1,0), (-1,0), (0,1), (0,-1)$, and write the unknowns as

$$x_0 = f(0,0), \quad x_1 = f(1,0), \quad x_2 = f(-1,0), \quad x_3 = f(0,1), \quad x_4 = f(0,-1).$$

Writing the harmonic condition at each point of R, replacing boundary values by β, and moving them to the right-hand side gives:

$$\begin{aligned}
(0,0): &\quad -4x_0 + x_1 + x_2 + x_3 + x_4 = 0,\\
(1,0): &\quad x_0 - 4x_1 = -\big(\beta(2,0) + \beta(1,1) + \beta(1,-1)\big) = -1,\\
(-1,0): &\quad x_0 - 4x_2 = -\big(\beta(-2,0) + \beta(-1,1) + \beta(-1,-1)\big) = -1,\\
(0,1): &\quad x_0 - 4x_3 = -\big(\beta(1,1) + \beta(-1,1) + \beta(0,2)\big) = -3,\\
(0,-1): &\quad x_0 - 4x_4 = -\big(\beta(1,-1) + \beta(-1,-1) + \beta(0,-2)\big) = 0.
\end{aligned}$$

In matrix form, $LX = B$ with

$$L = \begin{bmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -4 & 0 & 0 & 0 \\ 1 & 0 & -4 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 \\ 1 & 0 & 0 & 0 & -4 \end{bmatrix}, \qquad X = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ -1 \\ -1 \\ -3 \\ 0 \end{bmatrix}.$$

Solving. The last four equations express $x_1, \dots, x_4$ in terms of x_0:

$$x_1 = \frac{x_0 + 1}{4}, \qquad x_2 = \frac{x_0 + 1}{4}, \qquad x_3 = \frac{x_0 + 3}{4}, \qquad x_4 = \frac{x_0}{4}.$$

Substituting these into the first equation gives

$$-4x_0 + \frac{4x_0 + 5}{4} = 0 \quad \Longrightarrow \quad -3x_0 + \frac{5}{4} = 0 \quad \Longrightarrow \quad x_0 = \frac{5}{12}.$$

Therefore

$$f(0,0) = \frac{5}{12}, \qquad f(1,0) = f(-1,0) = \frac{17}{48}, \qquad f(0,1) = \frac{41}{48}, \qquad f(0,-1) = \frac{5}{48}.$$

Check. At $(0,0)$, the average of the four neighbors is $\frac{1}{4} \cdot \frac{17+17+41+5}{48} = \frac{1}{4} \cdot \frac{80}{48} = \frac{5}{12}$, as required. As one would expect, the values are larger near the top of the region, where $\beta = 1$, and they all lie between 0 and 1, the extreme values of β.

Part (b)

Theorem (Maximum Principle)**.** *Let f be a real-valued function on $\overline{R}$ that is harmonic at every point of R. Then the maximum value of f on $\overline{R}$ is attained at a point of ∂R. Likewise, the minimum value of f on $\overline{R}$ is attained at a point of ∂R.*

Proof. Since $\overline{R}$ is a finite set, f attains a maximum value M on $\overline{R}$. If M is attained at a point of ∂R, there is nothing to prove, so suppose $f(p) = M$ for some $p = (u, v) \in R$.

Step 1: a maximum at a point of R spreads to its neighbors. Let $q \in R$ with $f(q) = M$, and let q_1, q_2, q_3, q_4 be its neighbors. By the remark above, these lie in $\overline{R}$, so $f(q_i) \le M$ for each i. Since f is harmonic at q,

$$M = f(q) = \frac{f(q_1) + f(q_2) + f(q_3) + f(q_4)}{4}.$$

If some $f(q_i) < M$, then the right-hand side would be less than $\frac{4M}{4} = M$, a contradiction. Hence $f(q_i) = M$ for all four neighbors.

Step 2: walking to the boundary. Consider the points $(u + j, v)$ for $j = 0, 1, 2, \ldots$. Since R is finite, not all of these lie in R. Let $k \ge 1$ be the smallest integer with $(u + k, v) \notin R$. Then $(u + j, v) \in R$ for $0 \le j \le k - 1$, and $(u + k, v)$ is at distance 1 from $(u + k - 1, v) \in R$, so $(u + k, v) \in \partial R$.

We show by induction that $f(u + j, v) = M$ for $0 \le j \le k$. For $j = 0$ this holds by assumption. If $f(u + j, v) = M$ for some $j \le k - 1$, then $(u + j, v) \in R$, so Step 1 applies, and its neighbor $(u + j + 1, v)$ also has $f(u + j + 1, v) = M$.

In particular, $f(u + k, v) = M$ with $(u + k, v) \in \partial R$. So the maximum is attained on the boundary.

The minimum. The harmonic condition is linear in f, so $-f$ is also harmonic at every point of R. The maximum of $-f$ is attained at some point $b \in \partial R$, and then the minimum of f is attained at b. □

As a consequence, if f is harmonic on R, then for every $p \in \overline{R}$,

$$\min_{\partial R} f \;\le\; f(p) \;\le\; \max_{\partial R} f.$$

Part (c)

Theorem. *For every nonempty discrete region R and every function β on ∂R, the discrete Dirichlet problem has exactly one solution.*

Proof. Let n be the number of points in R.

The system is square. There is one unknown x_{uv} for each point of R, and one equation, the harmonic condition, for each point of R. So $LX = B$ is a system of n equations in n unknowns, and L is an $n \times n$ matrix. Moreover, L depends only on R, not on β. The boundary values enter only through the right-hand side B: the entry of B for a point $p \in R$ is minus the sum of β over the neighbors of p that lie in ∂R. In particular, if $\beta = 0$ then $B = 0$.

Solutions of the Dirichlet problem correspond exactly to solutions X of $LX = B$. Given X, set $f = x_{uv}$ on R and $f = \beta$ on ∂R. Then f is harmonic on R precisely when $LX = B$.

The homogeneous system has only the trivial solution. Suppose $LX = 0$. Then the corresponding function f, with $f = x_{uv}$ on R and $f = 0$ on ∂R, is harmonic at every point of R. By the maximum principle in Part (b), for every $p \in R$,

$$0 = \min_{\partial R} f \;\le\; f(p) \;\le\; \max_{\partial R} f = 0.$$

Hence $f(p) = 0$ for every $p \in R$, that is, $X = 0$.

Conclusion. For a square matrix L, the following are equivalent: L is invertible, and the homogeneous system $LX = 0$ has only the trivial solution. (This is the characterization of invertible

matrices from Chapter 1.) Therefore L is invertible. For any β, the system $LX = B$ then has the solution $X = L^{-1}B$. This solution is unique, because if $LX = B$ then $X = L^{-1}LX = L^{-1}B$.

Thus the discrete Dirichlet problem has exactly one solution for every boundary function β. $\square$